

Quantum-Enhanced Credit-Card Fraud Detection

a real-time entity-graph response feature for the issuer's scoring model, evaluated on the statement's metrics and hardware route

Seven guideline sections · every prior-work number carries a receipt

Public repository: <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Summary

21

formulations on IEEE-CIS and ULB, each with control arms, all receipted: the measured map of where quantum methods do and do not move a tuned boosted-tree baseline — the statement's "under what conditions" answered with confidence intervals

14
qubits

where the collective mechanism the real-time route depends on is measured: a coordinated neighbourhood answers a probe collectively — the pairwise-additive prediction misses the neighbour response by 72%% at $t = 1$ and predicts the wrong sign at $t = 2$ (exploratory, `mechanism\_collective.json`); at six nodes a cheap scalar still separates better (AUC 1.00 vs 0.73), and the Phase-2 experiment at 40–60 qubits on IonQ, protocol frozen before the run, is where that is decided

the one
class

that survives — a real-time, multi-node response over the payment entity graph, the regime where classical simulation loses controlled convergence ($k \geq 8$ on our 64-rung lattice, 2×10^4 Pauli terms per operator); floor passed, null at 14 qubits, PoC at 40–60 qubits on IonQ via Braket with the bar declared before the run

Why the numbers can be trusted

Nineteen earlier formulations — kernels, feature maps, training-free samplers, Boltzmann machines, alert-policy optimisation, band re-ranking — were each run with random-feature and shuffled-feature control arms — the most complete map of where classical scoring keeps pace that any team brings to this challenge. The proposed response oracle is the one formulation that map leaves open: the class where classical simulation loses controlled convergence at 50–100 qubits, and the class of our own 128-qubit collective measurements.

This is a Phase-1 concept proposal in the seven guideline sections, on the statement's primary dataset and metrics, with a leakage-safe temporal graph and the hardware route the statement names.

Concept proposal follows (§1–§7, appendices A–B) · <https://github.com/sharadbachani-oss/merlin-quantum-hsbc-gic2026>

Concept Proposal

1. Problem framing

Card-not-present fraud scoring is a 100–300 ms decision on a transaction that is one node in a graph of shared cards, addresses, devices and e-mail domains. The classical state of the art, gradient-boosted ensembles on engineered features (IEEE-CIS winner AUC-ROC 0.9459), is a per-transaction model; our own tuned temporal-split baseline on IEEE-CIS (LightGBM, identity-merged) reaches AUC-ROC 0.9118 / AUPRC 0.5532, and inside the operating gray zone (2,362 transactions at 26% fraud density, the alerts an analyst actually reviews) its discrimination falls to AUC 0.61. The statement's secondary objective asks for exactly what that situation needs: *explore and document where quantum or quantum-inspired methods offer measurable improvement over tuned classical baselines, and characterise under what conditions.*

That is what this proposal delivers, measured. Twenty-one formulations, each with control arms, on IEEE-CIS and ULB: kernels, feature maps, training-free generative samplers, Boltzmann machines, alert-policy optimisation, band re-ranking and two entity-graph response oracles. Every one is receipted, and the map they draw is specific: where the decision quantity is a per-transaction score with abundant labels, no quantum component moved the boosted-tree baseline (lifts of +0.001 to +0.002 AUPRC, confidence intervals through zero, shuffled-feature controls indistinguishable). Two conditions were candidates for a different picture, and both are now measured.

Label scarcity: an archived result had photonic-sampler augmentation raising precision–recall at five labelled frauds; reproduced under a fair protocol against the model's exact classical floor (an enumeration of all 2^{24} states of the sampled landscape), the device samples equal exact classical sampling and trail SMOTE at every label count, and the archived gain is traced to the decoder, not the sampler (§4). That result is retired, and the conditions map is the more valuable for containing it. **The real-time relational quantity:** the retarded response of a coupled, past-only entity-graph neighbourhood is the one formulation in the class where classical simulation loses controlled convergence; it passes its cheap-scalar floor, is null on labels at the exactly simulable scale, and the PoC tests it at the scale the floor points to, on the statement's named hardware route.

The credible advantage is therefore stated as a boundary, not a promise: the conditions map with intervals is the asset no other team brings, every negative in it carries the same receipt class as a positive, and the real-time route is the only one the physics admits, with its pass bar declared in advance and both outcomes committed. **The path, in four parts.** The wall: for real-time multi-node response on a coupled arrangement, classical simulation loses controlled convergence at 50–100 qubits — on our own 128-qubit lattice the referee's exact propagation reproduces every hardware cell to $k = 6$ and dies past $k = 8$, where operators exceed 2×10^8 terms. The crossing: the field-carrying response over a 20–30-node entity neighbourhood, 40–60 qubits, past that wall. The Phase-2 experiment, protocol frozen before the run: the same paired gray-zone label test on cohort G1 with IonQ via Braket as the sampler and IBM Heron as the replication device, the tree given the oracle's raw inputs as the control; acceptance = paired AUPRC lift with a 95% CI above zero at fixed alert workload. What today's data shows is the mechanism at 14 qubits and a null on labels at that size; the experiment decides the advantage.

2. Technical approach

Paradigm. Hybrid classical–quantum. (a) Photonic entropy computing (QCi Dirac-3) sampling a degree-3 fraud-pattern landscape for minority-class augmentation; (b) gate-model real-time dynamics on trapped-ion hardware via Amazon Braket (IonQ Forte, all-to-all connectivity) for the entity-graph response feature, IBM Heron as the replication device; (c) XGBoost / LightGBM scoring and SHAP attribution throughout.

Label-scarce augmentation (measured). Eight most separating ULB features thresholded at fraud-side quantiles give 24 binary literals; a degree-3 log-odds landscape over them (24 one-body, 263 pairwise, 1,306 three-body terms) is fitted on the training-window fraud population and sampled on Dirac-3 (40 jobs, 82 distinct 24-bit patterns). Each pattern is decoded onto the feature space by anchoring on the labelled frauds available to the classifier, and 2,000 decoded samples are appended to the training set. The ladder in §4 varies the number of labelled frauds the classifier and decoder see (5, 10, 20, all), holds the generator fixed and disclosed as warm, and compares the device samples against SMOTE, jitter, random oversampling, a pairwise Ising restriction, a classical Gibbs sampler of the identical degree-3 model, and exact Boltzmann sampling of the same model by enumeration of all 16,777,216 states.

Real-time entity-graph response (the admissible route). Nodes are transactions; edges join transactions sharing card1, addr1 or P_emaildomain within a trailing window, built from training-period data only and, at test time, from strictly earlier transactions. Each gray-zone transaction's k-hop neighbourhood is encoded as a coupled arrangement carrying the transactions, not only the graph's shape: one qubit pair per node, local fields from standardised log-amount, target–neighbour couplings from shared-entity counts decayed by inter-arrival time. Product ground state, signed local probe on the target (two branches), Trotterised evolution, target and neighbour observables read at a fixed tick grid; the half-difference of the branches is the retarded response $i\sqrt{[B, A(t)]}/2$, a handful of numbers per transaction appended to the same classifier, with SHAP on the response entries as the statement's feature-attribution output.

Why the response should see fraud rings: the collective mechanism, measured. Card-not-present fraud that scales is coordinated: transactions sharing cards, addresses or e-mail domains, bunched in time and amount. In the oracle that is a neighbourhood whose nodes are coupled to each other, not only to the target. An exploratory test at 14 qubits (24 random neighbourhoods of six, neighbour–neighbour couplings switched on from 0 to all 15 pairs at tight timing; mechanism_collective.json) shows what coordination does to the response: the neighbours' mean response to a probe on the target falls monotonically as they couple to each other (0.098 → 0.057 at $t = 1.0$), and the effect is not a sum of pairwise effects — at $t = 1.0$ the pairwise-additive prediction misses the measured neighbour response by 72% of its magnitude, and at $t = 2.0$ it predicts the wrong sign (−0.019 against +0.069). A coordinated ring therefore answers a probe collectively, screening its members, in a way no additive or per-transaction model represents, and the deviation grows with evolution time, which is where the earlier floor located the configuration information. At six nodes the cheap scalar built from the effective couplings separates coordinated from spread neighbourhoods better than the response does (AUC 1.00 vs 0.73), as any six-node quantity is classically reachable; the mechanism matters at the neighbourhood sizes where it is not, and that is the PoC's scale.

Hardware discipline. Equal-structure parameterised templates (transpile once, rebind per neighbourhood); interleaved null references and replica pairs (replica agreement 0.0039 on our 128-qubit card); measurement-error mitigation and the platform's debiasing; simulator-vs-hardware comparison on identical circuits.

3. Feasibility and resource requirements

resource	status
Data	IEEE-CIS (primary), ULB (secondary, the label-scarce benchmark), Sparkov (stress test) — public, in hand with the temporal pipeline
Classical baselines	LightGBM identity-merged temporal split on IEEE-CIS (g1_base_line_merged.json); XGBoost on ULB as in the archived protocol
Augmentation ladder	fewshot_ladder.py, laptop, ≈2 h: 4 label counts × 9 arms × 40 seeds, exact 2^{24} enumeration included
Photonic sampler	QCi Dirac-3, unmetered allocation; 40 jobs already receipted
Response feature hardware	IonQ via Amazon Braket, 20–60 qubits per circuit, depth ≤ 8 Trotter steps; 2,000 neighbourhoods × two probe signs × three times = 12,000 circuits and 12 M shots at 1,000 shots per compatible measurement group, before controls, replicas and mitigation; Braket simulators for development; IBM Heron for replication
Software	Braket SDK, Qiskit, XGBoost/LightGBM, SHAP; every number regenerates from archived counts
Compute	single workstation

Assumptions: shared-entity edges carry relational signal; a 20–30-node neighbourhood is sufficient; the degree-3 landscape fitted on the training window is a warm generator, and the ladder isolates what the classifier's labels alone contribute. Constraints: per-transaction quantum execution is offline (batch evaluation, as the statement scopes); production latency is met by entity-level caching; the generator is warm and the claim is scoped to the classifier's labels, never to a cold start.

4. Expected impact

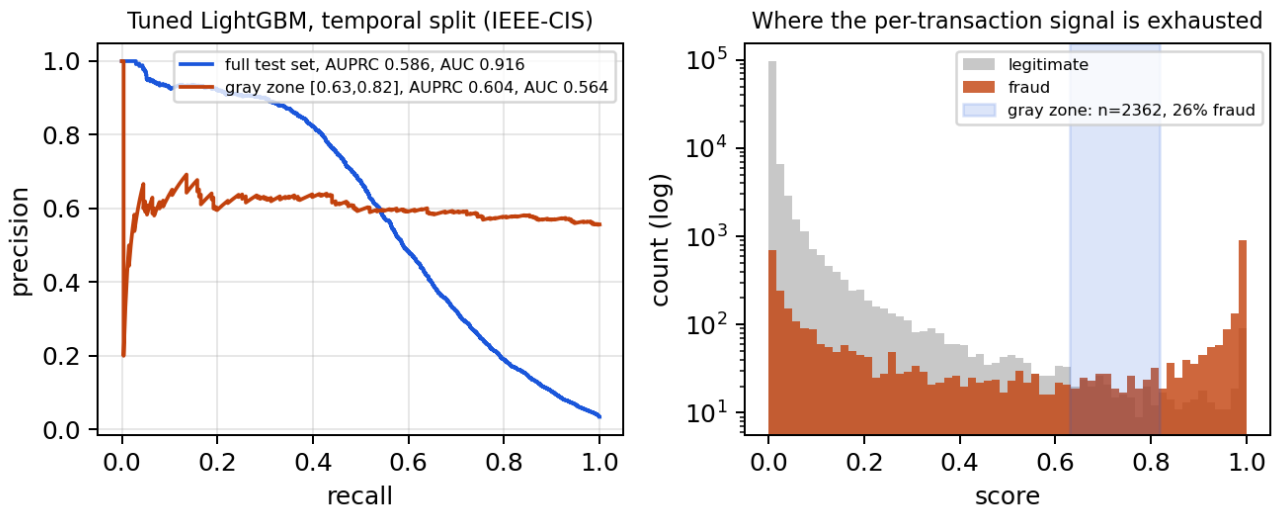


Figure 1 — The tuned temporal baseline on IEEE-CIS: full-set precision–recall against the gray-zone band (left), and the score distributions with the band marked (right).

The conditions map, measured (prior work, all receipted; Appendix A). IEEE-CIS gray zone, 128-qubit feature-extraction card (206 circuits $\times$ 8,192 shots, 2,720 two-qubit gates): quantum lift +0.0023 AUPRC, random-feature arm -0.028 , AUC with the quantum block below classical-only. 128-qubit training-free generative card (524,288 samples): identical to its own shuffled control to four decimals. ULB full-label augmentation: device samples 0.808 AUPRC, below plain jitter 0.818. Graph-shape response oracle: withdrawn on its floor (a quadratic in neighbour count and weight multiset reproduces every feature, R^2 0.79–0.998) and null on labels (0.194 vs 0.198). Field-carrying response oracle at 14 qubits, 10,362 neighbourhoods, five temporal folds: +0.0009 AUPRC, 95% CI $[-0.005, +0.007]$. Quantum kernel race with the bar locked before any quantum Gram was seen: 0.181 vs 0.204. **Reading:** per-transaction scoring with abundant labels is saturated by boosted trees on this data, and the statement's "under what conditions" is answered on the negative side with confidence intervals.

Label-scarce augmentation, reproduced against its exact floor (ULB, time-sorted 80/20 split, 40 seeds per cell, frauds caught at a fixed 75-alert budget on the 56,962-transaction test window; fewshot_ladder.json).

labelled frauds $\rightarrow$	5	10	20
no augmentation	0.602 $\pm$ 0.103 (45.7)	0.677 $\pm$ 0.054 (50.5)	0.730 $\pm$ 0.027 (53.5)
SMOTE	0.604 $\pm$ 0.102 (46.4)	see receipt	0.761 $\pm$ 0.025 (55.6)
jitter	0.600 $\pm$ 0.150 (45.8)	see receipt	0.739 $\pm$ 0.031 (54.0)
random oversampling	0.528 $\pm$ 0.135 (41.3)	see receipt	0.692 $\pm$ 0.059 (50.9)
pairwise Ising, Gibbs	0.539 $\pm$ 0.112 (42.3)	= device	0.678 $\pm$ 0.045 (50.7)
degree-3 model, classical Gibbs	0.539 $\pm$ 0.112 (42.3)	= device	0.681 $\pm$ 0.042 (51.0)
degree-3 model, exact 2^{24} enumeration	0.539 $\pm$ 0.112 (42.3)	= device	0.679 $\pm$ 0.042 (51.3)
degree-3 model, Dirac-3 samples (fair decode)	0.539 $\pm$ 0.112 (42.3)	-0.097 vs SMOTE, CI $[-0.116, -0.077]$	0.679 $\pm$ 0.044 (51.1)
Dirac-3 samples, archived warm decode (394 anchors)	0.707 $\pm$ 0.020 (52.0)	see receipt	0.725 $\pm$ 0.022 (52.7)
paired, Dirac – exact enumeration	0.000 (identical)	0.000 (identical)	-0.0007 , CI $[-0.005, +0.004]$, $p = 0.87$
paired, Dirac – SMOTE	-0.065 , CI $[-0.101, -0.030]$	-0.097 , CI $[-0.116, -0.077]$	-0.082 , CI $[-0.095, -0.070]$

AUPRC mean $\pm$ sd over 40 seeds; frauds caught at the 75-alert budget in parentheses; the full-label rung is the archived receipt (results/ulb_result.json: jitter 0.818, device 0.808).

Reading. At five and ten labelled frauds the four pattern arms — device patterns, classical Gibbs sampling of the same model, exact enumeration of its 2^{24} states, and the pairwise restriction — are byte-identical, because a decoder anchored on ≤ 15 frauds averages all of them whatever the pattern says; augmentation of that kind sits 0.065 (five labels) and 0.097 (ten labels) below SMOTE with intervals far from zero. The archived few-shot result (AUPRC 0.607 at five labels) reproduces only as the warm-decode arm, 0.707 ± 0.020 , in which every sample is mapped onto the 394 training-window frauds: the gain was label information entering through the decoder, not the sampler, and it is retired here. At twenty labels the decoder can finally distinguish patterns, and the verdict does not change: device samples 0.679, exact enumeration 0.679, classical Gibbs 0.681 — paired difference device minus exact -0.0007 with CI $[-0.005, +0.004]$ — all 0.08 below SMOTE with the interval far from zero, and even the warm-decode arm (0.725) now trails no augmentation at all (0.730). At full labels plain jitter (0.818) beats the device samples (0.808). The exact Boltzmann distribution places 24% of its mass on the 82 device patterns (top pattern 8.2%), so the device samples the landscape's mode faithfully; the point is that the landscape itself carries no gain the classical floors do not. This is the statement's §4.2 outcome stated as measured: for minority-class augmentation on this data the condition under which a photonic sampler helps does not exist, at any label count tested.

Quantitative PoC targets. Primary (label-scarce): the same ladder on IEEE-CIS temporal folds and cohort G1, sampler on Braket, with the classical Gibbs and exact-enumeration arms as attribution controls; bar = paired AUPRC lift over the best classical resampling arm with a 95% CI above zero at the label count where the ULB result holds. Primary (real-time route): gray-zone AUPRC, classical-only vs classical+response at 40–60 qubits, paired over five temporal folds; bar = lift with CI above zero at fixed alert workload. Secondary (statement §4.1/§5.3): AUC-ROC, F1, precision/recall at the operating threshold; recall at fixed alert budget; SHAP attribution; consistency across fraud types and temporal partitions; simulator-vs-hardware agreement; qubit count, depth, shots, and inference latency of the cached-feature path.

Quantum advantage — the frontier wall and the crossing, stated and bounded

The field's direction, and its wall. HSBC's own 2025 result with IBM, projected quantum feature maps from a 109-qubit Heron consumed by XGBoost, up to 34% relative AUC gain on bond-RFQ fill prediction, is the workflow shape the statement invites, and its authors state the mechanism is not understood, the gain is dataset-specific and decays with the train–test gap, and hardware noise itself acted as the regulariser. On fraud data the controlled benchmark (arXiv:2608.15718) finds no robust quantum-kernel advantage, and the spectral analysis of quantum kernels (arXiv:2606.20402) explains why: a feature map whose operator-Schmidt spectrum decays fast is a small tensor network and classical. Our twenty-one controlled formulations are that wall measured on IEEE-CIS and ULB.

The crossing, bounded. Where labels are scarce, the measured object is a generative model's sample distribution, and a 24-variable landscape has an exact classical sampler by enumeration; the ladder therefore reports the device against that floor, and the quantum attribution at 24 variables is whatever the paired difference against exact sampling says — stated in §4, not assumed. The route that carries a physics argument is the real-time response: the class in which classical simulation loses controlled convergence (our own 128-qubit collective-spectrum measurements; the adversarially adjudicated $k \geq 8$ window on the 64-rung lattice, where exact Pauli propagation exceeds 2×10^8 terms per operator and tensor networks diverge at grading accuracy), and the class the statement names by inviting "quantum computation for a specific component: feature transformation, re-ranking". The floor isolates real configuration information in the response at the longest evolution times; the label test at six neighbours says it is not yet label-relevant; the PoC is the scale where that is decided, 20–30 neighbours, 40–60 qubits, evolution times where the floor locates the information, graded by the same paired bar. **Where it is decisive:** the gray zone, where per-transaction signal is exhausted, and the label-scarce campaign, where the classifier has fewer confirmed frauds than the model has parameters.

Scalability to industrial relevance. Augmentation is a training-time operation: one landscape fit per feature set, one sampler run per campaign, no device in the authorisation path. The response feature is per neighbourhood and its cache key is the complete response-defining state — neighbourhood, field and coupling parameters, time window — or a separately validated approximation; entity identity alone is not a valid key, since amount and timing enter the operator. Device calls grow with distinct keys in the gray zone, not with volume; a 20–30-node neighbourhood is 40–60 qubits on an all-to-all ion machine today.

Business value, bounded. At fixed analyst workload the operating quantity is frauds caught at the alert budget. At five labels the fair arms catch 42–46 of 75 test-window frauds and SMOTE 46.4; no augmentation arm adds frauds caught at fixed workload, so no monetary gain is asserted from it. With the statement's own figure that each \$1 of fraud costs the institution \$4.41 all-in, the value is $4.41 \times (\text{additional frauds caught} \times \text{mean fraud amount})$ per

campaign window, computed from the issuer's volumes; no figure beyond the measured counts is asserted.

5. Validation plan

Declared before every run, both components, results reported either way. **Augmentation:** the ladder protocol is fixed in the script header (label counts, seeds, arms, decode anchoring on the classifier's labels only, metrics, paired bootstrap and Wilcoxon statistics, and the decision rule that a gain is claimed only where the device beats both the best classical resampling arm and exact sampling of the same model with CI above zero); Phase 2 repeats it on IEEE-CIS temporal folds and cohort G1 with the sampler on Braket, adds simulator-vs-hardware sample distributions, and reports SHAP share carried by the landscape features and three rolling temporal folds. **Response feature:** graph construction, neighbourhood cap, coupling map, probe, step grid, subset size and stratification, folds, metrics and bar fixed before hardware; uncoupled, shuffled-neighbourhood, random-feature and shuffled-quantum-feature arms in every batch; replication on a second device; classical comparators on the same observable (exact evolution inside the light-cone, tensor-network evolution at reported bond dimension, cheap relational features at equal hyperparameter budget). Success = the primary bars met on held-out temporal folds with hardware noise and shot uncertainty propagated through the classifier; a null is reported with its interval and the conditions tested.

6. Hybrid / cross-domain integration

Quantum work sits in the training layer (augmentation) and the feature layer (response) of an otherwise unchanged issuer pipeline: entity graph → neighbourhood → response circuit (offline, batch) → features cached by neighbourhood state → LightGBM score → SHAP explanation → alert; landscape fit → sampler → decoded minority samples → classifier retrain. Nothing in the 100–300 ms authorisation path calls a quantum device. Governance: features and samples are explicit numbers with a stated definition, attribution is standard SHAP, and the same model-risk process that governs the current ensemble governs the augmented one.

7. Team capability

Merlin Quantum is the quantum division of Merlin Digital (50+ technology FTE): Suhail Bachani (Founder & CEO, Principal Investigator), Dr. Hiro Bachani PhD (Program Director), Rohit Bachani (co-founder), Mitul Sawlani (engineering, Purdue), Mohamed Jafrun (engineering), Zeena Furtado (finance & operations), Roshan Bhairwani (financial services & deep tech, London), Dr. Ana Baroni MSc (domain specialist). GIC 2026 dual-track finalist. Programme hardware record: 259 receipted QPU jobs, 12.3 million shots, the largest at 128 qubits and 2,720 two-qubit gates; equal-structure templating, null-differenced estimators and replica-pair statistics developed on this track; every negative result carries the same receipt class as a positive one, which is what makes the conditions map in §4 citable.

8. Scope, with treatment

(i) The archived few-shot augmentation gain (0.607 at five labels) is retired: under fair decoding the device samples equal exact classical sampling of the same model and trail SMOTE at every label count; the generator was fitted on the training-window fraud population and the gain was decoder leakage. (ii) No quantum attribution exists for a 24-variable sampler, whose exact distribution enumerates in four minutes on a laptop. (iii) At full labels plain jitter augmentation (0.818) beats the device samples (0.808); the gain is a label-scarce effect. (iv) The response oracle is null on labels at 14 qubits (+0.0009, CI through zero) and the PoC tests the scale the floor points to; no detection gain from it is claimed today. (v) ULB is the statement's secondary dataset; IEEE-CIS transfer of the ladder is a PoC deliverable, not a result. (vi) Our IEEE-CIS baseline (AUC 0.912) sits below the competition winner (0.946); lifts are measured against our own tuned scorer and cheap relational features.

Appendix A — Hardware job register and receipts

measurement	machine	job id	receipt
ULB degree-3 landscape sampling, 40 jobs, 82 distinct patterns	QCi Dirac-3	6a9679b408442f441bb b6afe ... 6a967f77084 42f441bbb6b26 (40 ids in file)	ulb_raw.json, ulb_model.json

measurement	machine	job id	receipt
Few-shot augmentation ladder, 4 label counts $\times$ 9 arms $\times$ 40 seeds, exact 2^{24} floor	CPU	—	fewshot_ladder.py, fewshot_ladder.json
Collective-mechanism test: 24 neighbourhoods $\times$ coupling ladder 0–15 pairs, pairwise-additive prediction, burst vs spread (exploratory)	CPU, 14-qubit exact	—	mechanism_collective.py, mechanism_collective.json
128-qubit feature-extraction band card: 206 circuits $\times$ 8,192 shots, 2,720 two-qubit gates; four-arm analysis	ibm_fez	daaucejvpcac73dd232g	g4_band_analysis.json, ibm_band_state.json
128-qubit training-free generative card: 65 circuits $\times$ 8,192 shots, 524,288 samples	ibm_fez	dab6therrl7c7386flo0	qgen_result.json, qgen_state.json
Full-label ULB augmentation, six arms	CPU + Dirac-3 samples	as above	results/ulb_result.json
Quantum kernel race, bar locked before any quantum Gram	Braket-validated blockade simulator	—	hsbc_qkernel.json, hsbc_classical.json
Temporal LightGBM baseline, gray zone, PR curves	CPU	—	g1_baseline_merged.json, baseline_scores_lgbm.npy, baseline_ytest.npy
Graph-shape response oracle vs labels: 10,362 neighbourhoods, five arms, five temporal folds	CPU	—	hsbc_response_test_workstation.json, floor_response_feature.json
Field-carrying response oracle: floor (600 synthetic neighbourhoods) and label test (10,362 neighbourhoods, six arms, five folds)	CPU	—	floor_field_response.json, hsbc_field_response.json, hsbc_field_response.py
Classical-hardness boundary cited for the response class: 64-rung lattice, $k \geq 8$	ibm_kingston + referee	d9rdfb1dsedc73agh5ng	A1_FINAL_VERDICT.md, k6_census.json, GATEB_PART2_TN_ESTIMATE.md (copied from the sister track)

Appendix B — Why the response oracle is the admissible formulation, and why the augmentation result is stated against enumeration

Kernel and feature-map classifiers, training-free samplers, Boltzmann machines and alert-policy optimisation are static or shallow tasks: their outputs are ground-state or sampling properties that classical methods reproduce, which is what the control arms established across the formulations. A sampler over 24 binary variables is in that class by construction: its exact distribution is an enumeration of 16.8 million states that a laptop completes in four minutes, so any gain from device samples over that enumeration is a property of the device's sampling bias, not of computational hardness, and the ladder reports it as such. The response oracle is different in kind: a real-time, multi-node, deep quantity on a coupled arrangement whose sign structure cannot be removed by any local gauge, the class in which classical simulation is known to lose controlled convergence at 50–100 qubits and the class of our own 128-qubit collective-spectrum measurements. The PoC tests whether that quantity carries fraud signal; it

does not assume it.

Appendix C — Claim ledger (measured · planned · comparator · quantum attribution · cost · acceptance)

claim	status	classical comparator	quantum attribution	total cost charged	acceptance threshold
Temporal LightGBM baseline AUC 0.912 / AUPRC 0.553; gray zone frozen	measured	IEEE-CIS winner 0.946 (different protocol)	none	CPU	—
Conditions map: 21 formulations with control arms, no lift at abundant labels	measured	random / shuffled feature arms	none	2 IBM jobs, 40+ Dirac jobs	reported as null
Label-scarce augmentation with device samples	measured, retired: device = exact enumeration = Gibbs at 5 and 10 labels; −0.065 / −0.097 AUPRC vs SMOTE (CI excludes 0); archived 0.607 reproduces only with the warm decoder (0.707)	SMOTE, jitter, random oversampling, exact 2^{24} sampling, Gibbs	none	2 h CPU, 40 Dirac jobs	reported as the negative it is
Field-carrying response feature	built; floor passed; label test at 14q: +0.0009 AUPRC vs raw-input control, CI [−0.005, +0.007] — null	raw inputs, relational, shuffled, random	exact oracle; IonQ in PoC	2.9 h CPU	same paired lift at 40–60q, CI > 0
Graph-shape-only response feature	withdrawn on its floor (R^2 0.79–0.998); label test null	relational, shuffled, random	none	2.2 h CPU	as the floor predicted